Chem 150 **Answer Key** **Midterm** **Friday**

Part 1 (6 marks) **Multiple Choice Section**

For each statement below circle the letter that corresponds to the best answer. There is only one correct answer per question. **Each question answered correctly is worth 2 marks**.

1. Consider generating five mixtures of substances a, b, c, d and e in water. In each case enough water is added to 1.00 g compound to generate 1.000 L of mixture. In what mixture would the osmotic pressure be highest? Assume 25 °C.

This is an Osmotic pressure problem: $\Pi = MRT$. The solution with the greatest Molarity of solute will have the highest osmotic pressure. So wee need to convert mass of solute to moles of solute for each and see what happens to the solute when it is dissolved in water. Since Osmotic Pressure is a colligative property it will depend only on the quantity of solute, not its identity.

solute	mass	moles	Moles of aqueous "active" solute	Molarity of solute
CH_3-OH	1 g	0.03125 mol	0.03125 mol	0.03125 M
NaCl	**1 g**	**0.0171 mol**	**0.0171 mol + 0.0171 mol**	**0.0342 M**
S_8	1 g	0.004 mol	0 mol (insoluble in water)	0 M
$BaCl_2$	1 g	0.0048 mol	0.0048 mol + 0.0096 mol	0.0144 M
SO_2	1 g	0.0156 mol	0.0156 mol	0.0156 M

a) CH_3OH (l) → CH_3OH (aq) 32 g/mol, M = 0.03125 $\frac{mol}{L}$ CH_3OH (aq)

b) **<u>NaCl (s)</u>** ↔ Na^+(aq) + Cl^-(aq) 58.45 g/mol, M_{Na+} = 0.0171 $\frac{mol}{L}$, M_{Cl-} = 0.0171 $\frac{mol}{L}$

c) S_8(s) → S_8(s) non-polar, insoluble in polar water, does not contribute to osmotic pressure!

d) $BaCl_2$(s) ↔ Ba^{2+}(aq) + 2Cl^-(aq) 208 g/mol, M_{Ba2+} = 0.0048 $\frac{mol}{L}$, M_{Cl-} = 0.0096 $\frac{mol}{L}$

e) SO_2(g) ↔ SO_2(aq) (if all was actually dissolving, in reality only 2 % wt)

Since both the chloride and sodium ions contribute to the osmotic pressure the solute ion concentration entered in the equation π=MRT is higher for b) with 2 x 0.0171 M = 0.0342 M

2. Consider the substances a to e again. Which substance would form a heterogeny mixture with water?

a) CH_3OH (l)

b) NaCl (s)

c) **<u>S_8(s)</u>** not water soluble (non-polar eight-membered ring, no electronegativity difference)

d) $BaCl_2$(s)

e) SO_2(g)

3. In which of the substances a to e can the CHEMICAL BONDING not be explained by metallic, polar covalent or ionic bonding?

a) CH_3OH (l) polar covalent

b) NaCl (s) ionic

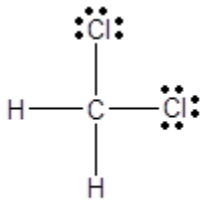
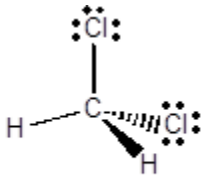
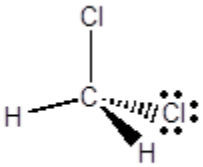
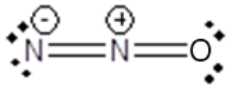
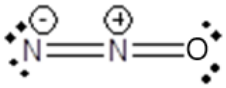
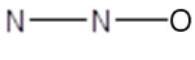
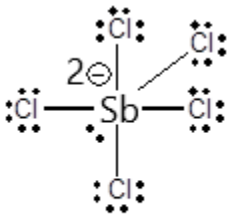
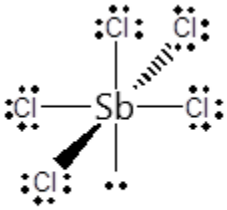
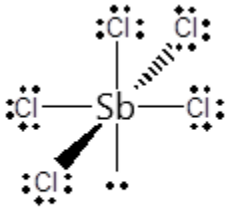
c) **<u>S_8(s)</u>** covalent (non-polar covalent)

d) $BaCl_2$(s) ionic

e) SO_2(g) polar covalent

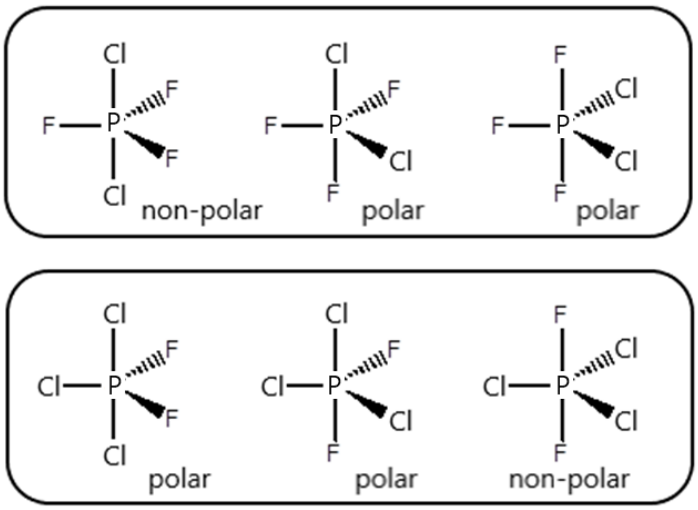
Part 2 (45 marks) Lewis structures, VSEPR and periodic properties and short answer questions

1.(**21 marks**) Identify the electron group geometry (EGG) and the molecular shape of the species below. Start by writing a valid Lewis structure and state the number of electron groups (#EG) around the central atom. Be sure to show all electrons and indicate any non-zero formal charges. Continue by drawing the EGG placing lone pairs and bonding pairs in the appropriate positions. **Name the EGG**. Next draw the molecular shape. Here multiple bonds, formal charges on atoms and lone pairs might be omitted for clarity. **Name the molecular shape**. Finally, indicate if the molecule or ion is polar or not. Please refer to the example below (SO_2) if in doubt about any of the instructions.

Lewis Structure	EGG	Parent shape	molecular shape	polar (Yes or No)
	4	 tetrahedral	 tetrahedral	Yes
	2	 linear	 linear	yes
	6	 octahedral	 square pyramidal	yes

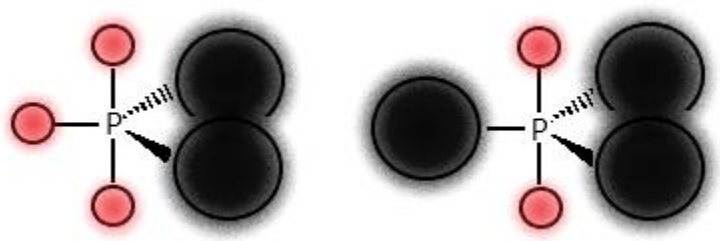
2.(**6 marks**) Experimental evidence shows that PF_3Cl_2 has a dipole moment whereas PF_2Cl_3 does not. Determine the structure of the two molecules and give a three-dimensional drawing. In a sentence or two elaborate on how the structures minimize overall repulsion and explain the experimental evidence.

Both molecules have trigonal bipyramidal shape. Let's work through the possibilities in which 2 Cl and 3 F atoms can be arranged around the central phosphorus (top frame below). The two Cl could be axial, which would create a fully symmetrical molecule in which all the bond dipoles would cancel out creating a non-polar molecule. This is not what the experimental evidence suggests! PF_3Cl_2 is polar. The other two possibilities (one Cl axial and one equatorial, as well as both Cl equatorial) would create polar molecules.



The second frame (above) shows the isomers of PF_2Cl_3. The only isomer that would cancel all bond-dipoles would be the one with all 3 Cl atoms in equatorial positions.

This makes sense since we learned that **lone-pairs in trigonal bipyramidal EGG always go in the equator as they require more space and repulsion is greatly reduced in the equator** (only two 90 ° neighbours, as suppose to the axial positions with three 90 ° neighbours). **Chlorine atoms are larger than fluorine atoms and would prefer the equatorial positions**, just like lone-pairs would. The structures that would be in agreement with these assumptions and the experimental evidence are shown below.



PF_3Cl_2 (polar) repulsions minimized and PF_2Cl_3 (non-polar) repulsion minimized,

3. (**2 marks**) Using the periodic table arrange the following in the expected order of increasing size:

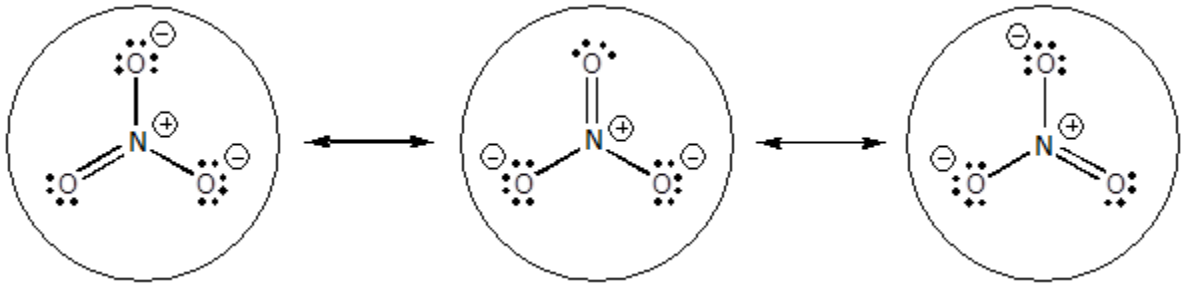
Cl, Rb, Ca, Si

Cl < Si < Ca < Rb

4.(**8 marks**) Complete the table below by naming four types of intermolecular forces. Give a representative example for each type of intermolecular force,

Type of intermolecular force	Example
Ion-dipole force	**Sodium chloride dissolved in water**
London forces (dispersion forces, also called induced dipole- induced dipole forces)	**Methane, argon, oil**
Dipole-dipole forces	**Forces between two HCl molecules**
Hydrogen bonding	**water**

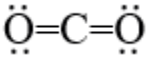
5. (**8 marks**) Draw all valid Lewis structures for the nitrate ion $[NO_3]^-$ Comment if the structures you have drawn are equivalent or not. State if the nitrate ion is polar or not.

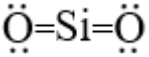


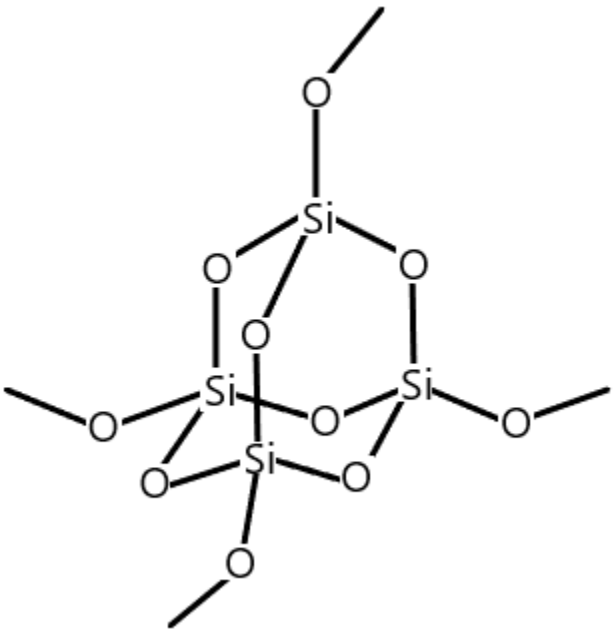
All three resonance structures are equivalent. The nitrate ion is non-polar (by resonance, resonance hybrid is non-polar)

Part 3 (15 marks) Intermolecular interactions, Colligative Properties and Gases and more short answer questions

1. (**5 marks**) The compounds CO_2 and SiO_2 have very different properties. CO_2 is a gas under standard conditions whilst SiO_2 is a solid with a melting point just under 2000 K. Explain the differences. Use arguments involving chemical bonding and intermolecular forces. Draw Lewis structures of both compounds that are consistent with their physical properties.

   Hard Question! We are already familiar with the Lewis structure of CO_2 

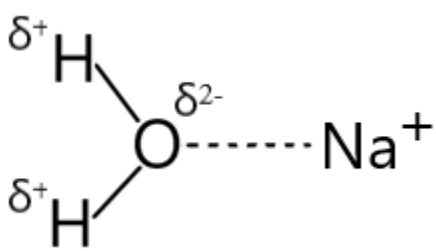
   The Lewis structure of SiO_2 must be different to account for the very different physical properties!  would also be a non-polar relatively small molecule with slightly larger dispersion forces. This would elevate melting and boiling points but perhaps by some 50 to 100 K, not by over 1800 K. What other options do we have? Oxygen can meet its two-bond requirement by making a double bond or **two single bonds**. Similarly, silicon can make two double bonds or **four single bonds**. In SiO_2 each silicon forms four single bonds to four oxygens with each oxygen bridging two silicon atoms. The result is a covalent network solid, the mineral quartz, of which a small structural segment is shown below:



2. (**3 marks**) How many electrons, protons and neutrons are there in an atom of ^{28}**Si**?

Number of Electrons	**14**
Number of Protons	**14**
Number of Neutrons	**14**

3. (**4 marks**) Draw a water molecule that are engaged in an ion-dipole interaction with a sodium ion. Be sure to draw the molecule in its correct geometry and align the molecule and the ion properly. Draw covalent bonds as solid lines (as usual) and the ion-dipole bond as dashed line.



4. (**3 marks**) Calculate the molar solubility of nitrogen in water that has been pressurized with 12 atm of air (78 % vol nitrogen) at 25 °C. The Henry's Law constant for nitrogen at that temperature is 6.0 x 10^{-4} $\frac{mol}{L\ atm}$.

$\mathbf{S_g = kp_g}$ = 6.0 x 10^{-4} $\frac{mol}{L\ atm}$ x 0.78 x 12 atm = 5.6 x 10^{-3} $\frac{mol}{L}$

Part 4 (19 marks) Numerical problems

1. (**5 marks**) The vapour pressure of liquid mercury is only 0.261 Pa at 25 °C. Nevertheless, dangerous levels build up in air that is in contact with mercury, especially in areas that are not ventilated well.

Calculate the concentration of mercury in air saturated with mercury vapour at 25 °C. State your answer in mg of Hg per cubic meter of air and state if it is safe to work in an environment saturated with Hg(g) given that the legal exposure limit is 0.1 mg/m^3 over an 8-hour work day whilst the recommended airborne exposure limit is 0.05 mg/m^3 over 10 hr work day.

Strategy:

Use the vapour pressure of mercury to find the moles on Hg(g) in 1 m^3 of air and convert that to mg/m^3.

pV = nRT and $n_{Hg(g)} = \frac{pV}{RT} = \frac{(0.261\ Pa)(0.001\frac{Kpa}{Pa})(1000\ L)}{\left(\frac{8.314 LkPa}{molK}\right)(298\ k)}$ = 1.0535 x 10^{-4} mol Hg(g) in 1000 L of air at saturation.

1.0535 x 10^{-4} mol/m^3 x $\frac{200.6\ g}{mol}$ = 0.0211g/m^3 = 21.1 mg/m^3

The air is not safe!

2.(**6 marks**) Magnesium can be uses as a "getter" in tungsten light bulbs to remove residual amounts of oxygen and nitrogen which would shorten the lifetime of the tungsten wire because it forms magnesium oxide and magnesium nitride respectively. Write a balanced equation for the reaction of nitrogen with magnesium. Assume a residual nitrogen partial pressure pN_2 of 3.45 x 10^{-4} kPa in a light bulb of 0.320 L volume at 20°C (and no residual oxygen). What mass of magnesium would react?

This question was directly from Problem Set 4 (gasses). See Answer Key Problem Set 4 for solution.

3.(**8 marks**) Many people and governments around the world are concerned about increasing CO_2 emissions into our shared atmosphere and huge efforts are underway to curb those emissions. The Mauna Loa Observatory on the big Island of Hawaii has been rigorously monitoring atmospheric CO_2 concentrations since 1967. The average monthly measurement for January 2023 was 419.48 ppm whilst the monthly average for January 2024 was 422.80 ppm (e.g. 422.80 mol of CO_2 in 1,000,000 mol of dry air). Estimate the annual increase of CO_2 in metric tons in our global atmosphere. Recall that pressure is force per area and force is mass times acceleration, which is 9.81 $\frac{m}{s2}$ on earths surface. Recall the surface of a sphere is A =4πr^2 and Earth's radius is 6371 km. Assume a composition of the air that is 78 % vol N_2, 21 % vol O_2, 1 % vol Ar. Assume standard temperature and standard pressure. Neglect water vapour and mountain ranges. Depending on your approach not all the data given might be needed.

Some of you started with pV = nRT and estimated the volume of the atmosphere using a certain height and got pretty close to a good estimate. We can avoid the Volume of the atmosphere altogether by starting with atmospheric pressure at sea-level. Since pressure is force per area and force is mass times acceleration, we can calculate the mass of the atmosphere. Next, we can convert mass to moles using an average molar mass for air (calculated from the abundance data) and finally use the difference in mol fraction between 2023 and 2024 to calculate the increase in the number of moles of CO_2 in the atmosphere and finally convert that to mass.

Average pressure at sea-level is 101.3 kPa

p = $\frac{F}{A} = \frac{ma}{A}$ We can calculate the mass of the atmosphere in two steps by calculating the mass per 1m^2 first and then expanding the area to the whole surface of earth. We can also do it all in one step by inserting the surface of earth in the equation. I often prefer working in steps as it gives more opportunity to check if intermediate results are sensible. This is especially important with road map question like this one where we do a lot of synthesis and are likely faced with having to evaluate very large (almost inconceivably large) numbers.

Mass of atmosphere per 1 m^2: m = $\frac{pA}{a} = \frac{(101300\ Pa)(1\ m2)}{9.81 m/s2}$ = **10326 kg** [Pa = N/m^2 and N = (kg x m)/s^2]

Let's evaluate this first answer. Recall that pressure of the atmosphere can also be expressed in mmHg

1 atm = 760 mm Hg (formula sheet) or 10 m of water (Otto von Guericke's barometer we talked about in class, Magdeburg halfspheres slide). What does this mean? The mercury in the mercury barometer rises 760 mm as the force of a column of Hg 760 mm high is equal of to force exerted by the mass of the atmosphere. A column of mercury, 760 mm high (or water 10 m high) is thus equal in mass to a column of atmosphere over the same area. Mercury is about 13 times denser than water (10 m/ 0.76m = 13) Envision a column of water 10 m tall by stacking ten cubes of water each measuring 1m x 1m x 1m on top of another. The mass of this tower of water would have a mass of 10 metric tons (10,000 kg) spread out over an area of 1 m^2. This lines up with our first result. Divers know that for every 10 m they descent the pressure increases by one atmosphere.

Now we can calculate the mass of the entire atmosphere by expanding the area from 1 m^2 to earths surface.

A = $4\pi r^2 = 4\pi(6371000\ m)^2 = 5.1 \times 10^{14}\ m^2$.

$m_{atmosphere}$ = **10326 kg/m^2 x** $5.1 \times 10^{14}\ m^2 = 5.25 \times 10^{18}$ kg = **5.25×10^{21} g**

How many moles is that? We need the molar mass of air which is a mixture of mainly three gasses. Let's do a weighted average:

$0.78 \times 28.013 \frac{g}{mol} + 0.21 \times 32.0 \frac{g}{mol} + 0.01 \times 40.0 \frac{g}{mol}$ = 29.0 g/mol

So the number of moles of air (or gas molecules) in the atmosphere is: n = $\frac{mass}{molar\ mass} = \frac{5.25\ x\ E21\ g}{29\ g/mol}$

$n_{air} = 1.8 \times 10^{20}$ mol

Now we can get to the increase of CO_2 in the atmosphere in the last year. It increased by 422.80ppm – 419.48 ppm = 3.32 ppm or 3.32 mol CO_2 per 1,000,000 mol dry air.

So expanding that mol fraction to the entire atmosphere gives an annual increase of:

$\frac{3.32\ mol}{1{,}000{,}000\ mol}$ x 1.8 x 10^{20} mol = **6×10^{14} mol CO_2** which can easily be converted to mass:

6.0×10^{14} mol CO_2 x 44.011 $\frac{g}{mol}$ = 27 x 10^{15} g = 27 x 10^{9} t

<u>Approximately 27 billion metric tons of CO_2 were emitted globally in 2023.</u>

One of your answers reduced this problem efficiently to two lines by calculating the increase in partial pressure of CO_2 directly from the increase in mole fraction. Thus, the increase in partial pressure of CO_2 is approximately:

$\frac{3.32\ mol}{1{,}000{,}000\ mol}$ x 101300 Pa = 0.336Pa which translates to a mass of:

m = $\frac{pA}{a}$ = $\frac{(0.336\ Pa)(5.1\ E14\ m2)}{9.81 m/s2}$ = 17.5 x 10^{12} kg = 18 Billion tons

This approach is low by about a third. Obviously assuming ideal conditions only stretches so far. It is nevertheless an amazingly efficient estimate.

Formulas and Constants for the Midterm

$pV = nRT$

$p_iV_i(T_i)^{-1} = p_fV_f(T_f)^{-1}$

$P_{total} = p_1 + p_2 + p_3 + ... + p_n$

$V_{mol} = 22.4\ L\ mol^{-1}$ at STP (1 atm = 101.325 kPa = 760 mm Hg and 273.15 K)

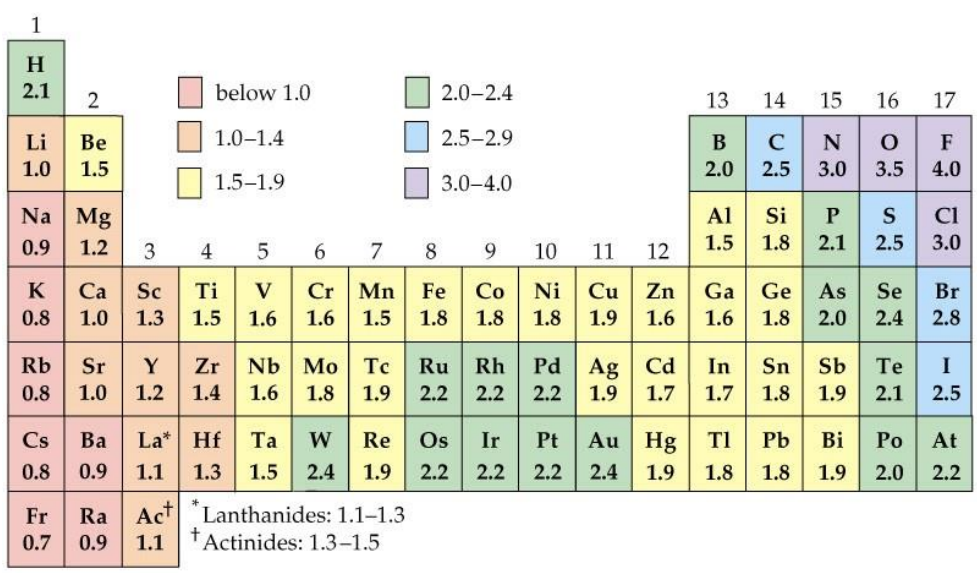
$R = 8.3145\ L\ kPa\ mol^{-1}\ K^{-1}$

Avogadro's number $N_a = 6.022 \times 10^{23}\ mol^{-1}$

Unified atomic mass unit $1\ u = 1.660 \times 10^{-27}\ kg$

$\Pi = MRT$

$P_A = x_AP°_A$

$S_g = kp_g$

Electronegativities of the elements